

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

BSc. (Engineering), Second-Semester Examination, August, 2023

First (1st) Year

COMPUTER ENGINEERING

EE 152: BASIC ELECTRONICS

TIME: 2hrs

PROGRAM OF STUDY:

INDEX NO:.....

Instructions

- i. Candidates are to indicate their Index Number and Program of Study in the spaces provided.*
- ii. Candidates are to submit ONLY the Scannable Sheet.*
- iii. Choose from the options lettered A to D, the most suitable answer to the question statements*
- iv. Shade the most suitable answer to the question statements on the Scannable Sheet provided*
- v. Candidates are to write the answer at the back of the Scannable Sheet if he/she thinks the right answer cannot be found in the options provided*
- vi. Each correct answer carries a mark*

**CANDIDATES ARE ALLOWED TO TAKE THE QUESTION PAPERS
AWAY**

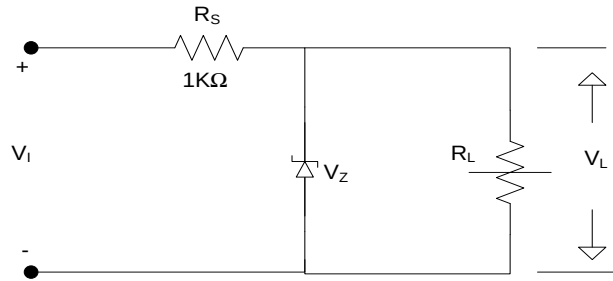


Fig .1.2:

The voltage regulator circuit of fig .1.2 is required to maintain the load voltage V_L at 6V. The circuit derives its power from a supply of $V_I = 30V$. The breakdown voltage of the zener diode, $V_Z = 6V$, and the maximum current that can be permitted to flow through the zener diode is $=12mA$.

1. Determine the minimum value of R_{Lmin} which will ensure this requirement.
(Take $I_{Zmin} = 10\%$ of I_{Zmax}).
 - a. 500Ω
 - b. 563Ω
 - c. 263Ω
 - d. 632Ω
2. Determine the maximum value of R_{Lmax} which will ensure this requirement.
 - a. 500Ω
 - b. 563Ω
 - c. 263Ω
 - d. 632Ω
3. Determine the minimum value of the power drop across the zener diode.
 - a. $2.350W$
 - b. $0.3156W$
 - c. $3.156W$
 - d. $4.012W$
4. Determine the maximum value of the power drop across the zener diode.
 - a. $2.350W$
 - b. $0.3156W$
 - c. $3.156W$
 - d. $4.012W$
5. In a semi-conductor diode, the barrier offers opposition to
 - a. holes in P-region only
 - b. free electrons in N-region only
 - c. majority carriers in both regions
 - d. majority as well as minority carriers in both regions
6. In an ideal diode there is no breakdown, no _____ current, and no forward _____ drop.
 - a. reverse, voltage
 - b. forward, current
 - c. forward, voltage
 - d. reverse, current

7. For a P-N diode, the number of minority carriers crossing the junction depends on
 - a. forward bias voltage
 - b. potential barrier
 - c. rate of thermal generation of electron hole pairs
 - d. none of the above
8. In a half wave rectifier, the load current flows
 - a. only for the positive half cycle of the input signal
 - b. only for the negative half cycle of the input signal
 - c. for full cycle
 - d. for less than fourth cycle
9. Silicon is not suitable for fabrication of light emitting diodes because it is
 - a. an indirect band gap semiconductor
 - b. direct band gap semiconductor
 - c. wideband gap semiconductor
 - d. narrowband gap semiconductor
10. When avalanche breakdown occurs covalent bonds are not affected.
 - a. True
 - b. False
11. If too large current passes through the diode
 - a. all electrons will leave
 - b. all holes will freeze
 - c. excessive heat may damage the diode
 - d. diode will emit light
12. The ripple factor of a full-wave rectifier circuit compared to that of a half wave rectifier circuit without filter is
 - a. half of that for a half 'wave rectifier
 - b. none of the above.
 - c. equal to that of a half wave rectifier.
 - d. less than half that for a half-wave rectifier circuit
13. What is the necessary a.c. input power from the transformer secondary used in a half wave rectifier to deliver 500 W of d.c. power to the load?
 - a. 308 W
 - b. 848 W
 - c. 616 W
 - d. 1232 W
14. The RMS value of a half wave rectifier current is 10 A. Its value for full wave rectification would be
 - a. 10 A
 - b. $(20/\pi)$ A
 - c. 14.14 A
 - d. 20 A.

15. Peak inverse voltage for a diode is the
- voltage corresponding to rated maximum voltage
 - maximum voltage that can be applied across the diode in the conducting direction
 - maximum voltage that can be applied across the diode in the non-conducting direction
 - none of the above.
16. If a 169.7 V half-wave peak has an average voltage of 54 V, what is the average of two full-wave peaks?
- 119.9 V
 - 108.0 V
 - 115.7 V
 - 339.4 V
17. In general, LEDs operate at voltage levels from _____ V to _____ V.
- 1.0, 3.0
 - 1.7, 3.3
 - 0.5, 4.0
 - None of the above
18. For the same secondary voltage, the output voltage from a centre-tap rectifier is
- twice
 - thrice
 - four times
 - one half
19. A half-wave rectifier has an input voltage of 240 V r.m.s. If the step-down transformer has a turns ratio of 8 : 1, what is the peak load voltage ? Ignore diode drop
- 27.5
 - 86.5
 - 30
 - 42.5
20. Four diodes are used in a full wave rectifier circuit, the forward resistance which may be considered constant at 1ohms and infinite reverse questions. The alternating supply voltage is 240vrms and load resistance is 480, Find the mean load current
- 0.7
 - 0.45
 - 0.6
 - 0.8
21. A resistor has 10V across it and 1mA flowing through it. What is the resistance of the resistor?
- 1K
 - 100 Ohms
 - 1000 Ohms
 - 10K

22. What is the voltage drop across an ideal diode?
- a. 0.1V
 - b. 0.3V
 - c. 0.7V
 - d. 0.0V
23. Three silicon diodes are placed in series in the same direction. What is the overall voltage drop of the diodes?
- a. 0.7V
 - b. 1.9V
 - c. 2.1V
 - d. 2.2V
24. The importance of including a capacitor in a rectifier is to.....
- a. Filter out ripple voltage
 - b. Store charges
 - c. Discharge charges
 - d. Regulate output voltage
25. The Zener diode operates in the region
- a. Forward
 - b. Reverse-breakdown
 - c. Active
 - d. Saturation
26. The breakdown voltage of a zener diode can be controlled by.....
- a. Changing the type of semiconductor used
 - b. Controlling the voltage across the zener diode
 - c. Controlling the current flowing through the zener diode
 - d. Controlling the doping level of the zener diode
27. Zener diodes allow current flow in how many directions?
- a. One
 - b. Two
 - c. Three
 - d. Four
28. The major application of Zener diodes is in.....
- a. Voltage regulation
 - b. Rectification
 - c. Amplification
 - d. None of the above
29. In designing Zener diode circuits, a resistor is placed in series with the Zener diode to.....
- a. Control the amount of regulation
 - b. Limit the voltage across the diode
 - c. Limit the current flowing through the diode to safe levels
 - d. None of the above

30. Ama Vida designed a Zener diode regulator for a circuit she is building for her basic electronics project. Upon power on, the Zener diode heats up and gets damaged although the voltage supplied did not exceed the max voltage for the Zener diode. What is the most probable problem with the circuit?
- The input voltage is too low
 - The input voltage is too high
 - The series resistance is too high
 - The series resistance is too low
31. The voltage across a Zener diode remains constant after which situation?
- Reverse saturation
 - Breakdown
 - Conduction
 - Reaction
32. Kwame David is designing a feedback controlled low dropout voltage regulator. He needs a 4.3 voltage reference for his circuit. What type of Zener diode should he purchase for his circuit?
- A high power Zener diode
 - A low power Zener diode
 - A Zener diode with $V_Z = 4.3V$
 - A Zener diode with $V_Z = 0.7V$
33. For basic operation of a transistor the base-emitter junction is _____ biased.
- forward-
 - semi-
 - reverse-
 - not
34. For basic operation of a transistor the collector-base junction is _____ biased.
- semi-
 - not
 - reverse-
 - forward-
35. The output or collector characteristics for a common base transistor amplifier shows that as a first approximation the relation between I_E and I_C in the active region is given by _____.
- $I_E \ll I_C$
 - $I_E \approx I_C$
 - $I_E = I_C$
 - $I_E \gg I_C$
36. In the saturation region, the base-emitter junction _____.
- and the base-collector junctions are both forward-biased
 - is forward-biased while the base-collector junction is reverse-biased
 - and the base-collector junctions are both reverse-biased
 - is reverse-biased while the base-collector junction is forward-biased

37. In the cut-off region, the base-emitter junction _____.
 a. and the base-collector junctions are both reverse-biased
 b. is reverse-biased while the base-collector junction is forward-biased
 c. and the base-collector junctions are both forward-biased
 d. is forward-biased while the base-collector junction is reverse-biased
38. In the active region, the base-emitter junction _____.
 a. is forward-biased while the base-collector junction is reverse-biased
 b. is reverse-biased while the base-collector junction is forward-biased
 c. and the base-collector junctions are both reverse-biased
 d. and the base-collector junctions are both forward-biased
39. In a small-signal transistor, the typical range of the parameter α is _____.
 a. greater than 1
 b. almost equal to 1 but always less than 1 (0.9 to 1.0)
 c. almost equal to 1 but always greater than 1 (1.0 to 1.1)
 d. between 0 and 1
40. The common-base, short-circuit, amplification factor is better known as _____.
 a. dc β
 b. dc α
 c. ac α
 d. ac β
41. The common-emitter, forward-current, amplification factor is better known as _____.
 a. ac β
 b. dc α
 c. ac α
 d. dc β
42. In a small-signal transistor, the typical range of the parameter β is _____.
 a. large and in the range of about 50 to 400
 b. between 0 and 100
 c. almost equal to 100 but always less than 100 (90 to 100)
 d. greater than 100
43. A BJT has measured dc current values of $I_B = 0.1$ mA and $I_C = 8.0$ mA. When I_B is varied by 100 μ A, I_C changes by 10 mA. What is the value of the β_{ac} for this device?
 a. 800
 b. 80
 c. 100
 d. 10

44. A BJT has measured dc current values of $I_B = 0.1 \text{ mA}$ and $I_C = 8.0 \text{ mA}$. When I_B is varied by $100 \mu\text{A}$, I_C changes by 10 mA . What is the value of the β_{dc} for this device?
- 800
 - 100
 - 10
 - 80
45. When a BJT is operating in the saturation region the voltage drop from the collector to the emitter V_{CE} is approximately equal to _____.
- zero (about 0.3 Volts)
 - the emitter voltage
 - the collector supply voltage
 - the collector current times the collector resistor
46. When a BJT is operating in the active region, the voltage drop from the base to the emitter V_{BE} is approximately equal to the _____.
- diode drop (about 0.7 V)
 - base current times the base resistor
 - emitter voltage
 - base bias voltage
47. BJTs are commonly used as _____.
- series damper circuits
 - the primary components in rectifiers
 - the primary components in amplifiers
 - All of the above
48. V_{CE} is measured _____.
- from the collector terminal to the emitter terminal
 - from the emitter terminal to ground
 - from the collector-emitter junction to ground
 - None of the above
49. Why is the arrow on the BJT schematic symbol important?
- It identifies the collector terminal and the type of BJT.
 - It identifies the base terminal and the type of BJT.
 - It identifies the emitter terminal and the type of BJT.
 - None of the above/below
50. In most cases, which two of the three BJT terminal currents are approximately equal in value?
- emitter current and base current
 - collector current and base current
 - collector current and emitter current
 - All currents are approximately equal.

51. Which of the following biasing combinations is not normally associated with one of the three transistor operating regions'?
- a. E-B junction = reverse, C-B junction = forward
 - b. E-B junction = forward, C-B junction = reverse
 - c. E-B junction = reverse, C-B junction = reverse
 - d. All of the above/below
52. The condition where an increase in bias current will not cause further increases in collector current is called _____.
- a. cutoff
 - b. active operation
 - c. saturation
 - d. All of the above/below
53. β is the ratio of _____.
- a. collector current to base current
 - b. emitter current to collector current
 - c. collector current to emitter current
 - d. base current to collector current
54. A given BJT has an emitter current of 12 mA and a base current of 600 μ A. What is the value of β with subscript((dc))?
- a. 200
 - b. 21
 - c. 20
 - d. 19
55. A given BJT has an emitter current of 15 mA and a collector current of 14.95 mA. What is the exact value of β ?
- a. 250
 - b. 1.003
 - c. 299
 - d. 300
56. For a given BJT, $\beta = 400$. What is the value of α for the device?
- a. 0.9975
 - b. 0.002
 - c. 1.00
 - d. 1.0025
57. A given BJT has an alpha of 0.9985 and a collector current of 15 mA. What is the value of base current?
- a. 15 mA
 - b. 15.15 mA
 - c. 14.85 mA
 - d. None of the above

58. Which transistor amplifier configuration is the most commonly used?
- common-collector
 - common-base
 - common-emitter
 - None of these are used more often than the others.
59. A given transistor has ratings of maximum collector current equal to 200 mA and a beta that varies between 150 and 200. What is the maximum allowable value of base current for the device?
- 4 mA
 - 1 mA
 - 1.33 mA
 - None of the above/below
60. A BJT has measured dc current values of $I_B = 1$ mA and $I_C = 80$ mA. When I_B is varied by 100 μ A, I_C changes by 10 mA. What is the value of β_{dc} for the device?
- 10
 - 800
 - 80
 - 100
61. A BJT has measured dc current values of $I_B = 1$ mA and $I_C = 80$ mA. When I_B is varied by 100 μ A, I_C changes by 10 mA. What is the value of β_{ac} for the device?
- 800
 - 80
 - 100
 - 10
62. When a transistor is in saturation V_{CE} is approximately equal to_____.
- 0.3 Volts
 - collector current times collector resistor
 - emitter voltage
 - collector supply voltage
63. For the CB configuration breakdown occurs when _____ is too high but in the CC configuration too high a _____ results in breakdown.
- V_{CB} , V_{CE}
 - V_{BE} , V_{CE}
 - V_{CB} , V_{BE}
 - V_{BE} , V_{CE}
64. As _____ decreases the noise figure increases.
- source resistance
 - frequency
 - collector current

- d.all of the above/below
65. Moore's Law states that transistor count in an integrated circuit will double every _____ years.
- a.4
 - b.6
 - c.8
 - d.2
66. When a BJT is biased in the active region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.
- a.reverse; reverse
 - b.reverse; forward
 - c.forward; forward
 - d.forward; reverse
67. When a BJT is biased in the cut-off region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.
- a.reverse; reverse
 - b.reverse; forward
 - c.forward; reverse
 - d.forward; forward
68. When a BJT is biased in the saturation region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.
- a.reverse; reverse
 - b.forward; reverse
 - c.reverse; forward
 - d.forward; forward
69. When a BJT is biased in the cutoff region the collector-to-emitter voltage is typically equal to _____.
- a.0.03 V
 - b.the collector current times the collector resistor
 - c.the collector supply voltage
 - d.the emitter voltage
70. The term bipolar means:
- a.the use of both holes and electrons as carriers
 - b.polarity is unimportant
 - c.electrons make up the majority carriers
 - d.diodes only
71. In order for a transistor to operate properly:
- a.the (BE) junction is reverse-biased
 - b.(BE) and (BC) bias are always the same
 - c.the (BE) junction is forward-biased
 - d.(BC) bias is unimportant

72. Compared to the collector and emitter regions
- a. the base and emitter are always doped the same
 - b. the base region is lightly doped
 - c. the base region is doped the same
 - d. the base region is heavily doped
73. Regarding transistor operation in general:
- a. collector current controls base current
 - b. a small base current controls a large collector current
 - c. a large base current controls a small collector current
 - d. emitter current controls collector current
74. DC beta refers to:
- a. a transistor's gain characteristics
 - b. the ratio of emitter current to collector current
 - c. the ratio of base current to collector current
 - d. both a and b
75. With respect to a collector characteristics curve
- a. beta cannot be calculated from the curve
 - b. after (V_{ce}) reaches 0.7 volts collector current is close to its full value
 - c. only one base current can be represented on a characteristic curve
 - d. changing base currents have little effect on collector current
76. Name the condition whereby collector current reaches its peak.
- a. cutoff
 - b. Beta max
 - c. saturation
 - d. Beta min
77. A cutoff transistor has:
- a. $V_{CE} = V_{CC}$
 - b. V_{CE} cannot be calculated at cutoff
 - c. $V_{CE} = 0\text{ V}$
 - d. none of the above/below
78. $V_{CC} = 12\text{ volts}$, $R_C = 1\text{k}$, determine the load line I_C and V_{CE} intercepts
- a. $I_C = 1.2\text{mA}$ and $V_{CE} = 12\text{m volts}$
 - b. $I_C = I_B = 12\text{mA}$ and $V_{CE} = 12\text{m volts}$
 - c. $I_C = .012\text{mA}$ and $V_{CE} = 1.2\text{ volts}$
 - d. $I_C = 12\text{mA}$ and $V_{CE} = 12\text{ volts}$
79. What makes a base-biased circuit a good choice for switching applications?
- a. simple, single resistor bias circuit
 - b. calculate saturation and cutoff without considering beta
 - c. both a and b
 - d. none of the above/below

80. Collector feedback bias works on what principle?
- a. rising collector currents tend to attenuate base current
 - b. positive feedback adds to collector current
 - c. falling collector currents tend to attenuate base current
 - d. base current will always remain constant
81. Voltage divider bias provides a high degree of stability as long as what principle is observed?
- a. I_{R2} is ten times greater than I_B .
 - b. R_L is smaller than $R_1 + R_2$
 - c. $I_B = I_{R2}$
 - d. Either a or b
82. Which bias circuit is found chiefly in integrated circuit applications?
- a. voltage divider bias
 - b. base bias
 - c. collector feedback bias
 - d. emitter bias
83. Transistor data sheets provide what information?
- a. electrical characteristics
 - b. thermal characteristics
 - c. maximum ratings
 - d. all of the above/below
84. Coupling capacitors are used because
- a. they provide isolation between bias and signal sources
 - b. they provide an closed path for AC signals
 - c. they amplify inputs
 - d. they prevent feedback noise
85. With respect to the input signal of an amplifier
- a. the output is always in phase
 - b. the output is always out of phase
 - c. the base and collector signals are always in phase
 - d. the base and collector signal are always out of phase
86. In a Common-Emitter amplifier
- a. the emitter is always above ground potential
 - b. the output signal is taken from the emitter
 - c. the emitter is the reference for the input signal
 - d. the output is taken from the collector
87. How can we ensure that a Common Emitter amplifier has a high input impedance?
- a. always calculate the circuit parameters using a known B_{dc}
 - b. add a resistor that appears in series with the source
 - c. add a swamping resistor
 - d. all of the above/below

88. Regarding the Common Emitter amplifier
- a. the input is applied to the base
 - b. the output is at the emitter
 - c. the input and output is measured at the base
 - d. both a and b
89. Common-Collector amplifiers are preferred to drive low impedance loads because
- a. low input impedance buffers output
 - b. CC output impedance is low
 - c. CC amplifiers have high voltage gain
 - d. all of the above/below
90. Which amplifier configuration is the least often used?
- a. Common Base
 - b. Common Emitter
 - c. Darlington Pair
 - d. Common Collector
91. What is the primary difference between a Common Base and Common Emitter circuit?
- a. where the output is taken from
 - b. bias circuit
 - c. emitter ground
 - d. where the input signal is fed into the amplifier
92. A transistor in saturation is equivalent to what condition?
- a. an open switch
 - b. a closed switch
 - c. a or b depending on where voltage is measured
 - d. none of the above/below
93. If the base of a transistor is open, the voltage at the collector will be
- a. V_B
 - b. V_{CC}
 - c. $V_{CC}/2$
 - d. it depends on the beta of the transistor
94. If the collector of a transistor is open, the voltage at the emitter will be
- a. $V_B - 0.7 \text{ V}$
 - b. V_B
 - c. V_{CC}
 - d. $V_{CC}/2$
95. If V with subscript((CE)) is 13 V and V_{CC} is 15 V, this transistor is
- a. approaching saturation
 - b. a pnp transistor
 - c. approaching cutoff
 - d. properly biased

The diodes used in a bridge rectifier circuit have the following parameters:

$$V_T = 0V, \quad r_f = 15\Omega, \quad R_R = \infty, \quad V_Z = \text{very large}$$

The secondary winding of the transformer has a resistance $R_S = 0\Omega$. A load of 970Ω is connected across its output. If the voltage across the secondary is $v_s = 30 \sin \omega t$ volts,

Use the above information to solve questions 96 to 100

96. Determine the PIV rating of the diodes
- 40V
 - 30V
 - 60V
 - 25V
97. Determine the DC current through the load considering the voltage drop across the Diodes
- 0.06024A
 - 0.19720A
 - 0.01909A
 - 0.00147A
98. Determine the DC Voltage on the load considering the voltage drop across the Diodes
- 18.328V
 - 18.238V
 - 18.812V
 - 18.338V
99. Determine the DC power supplied to the load
- 0.3605W
 - 0.3591W
 - 0.3840W
 - 0.3025W
100. What is the conversion efficiency of this rectifier
- 0.7637pu
 - 0.7885pu
 - 0.7640pu
 - 0.7857pu